Functions, lists and algorithms: review



Programming · Level 3 · Unit check

In this unit

- Functions
- Lists and loops
- Searching
- Sorting

The big ideas

- Functions do one job, return a result and are tested with assert
- Lists count from 0; loops count, total and find the largest
- Binary search halves a sorted list: about 10 looks for 1,000
- Sorts differ in work; judge correctness first, with edge cases

Activity: Break my function

1. Warm-up (3 min): One idea from each lesson.
2. Write (7 min): A small function and three asserts.
3. Break (7 min): Swap; write an edge case that breaks it.
4. Fix (3 min): Fix it and rerun every assert.

Let’s discuss

1. Which algorithm surprised you most?
2. Which edge case broke the most code today?
3. When would you use a set instead of a sorted list?

Exit ticket

- One algorithm I can now write and test from memory